

Summary

PhD-trained Statistician and Data Scientist with expertise in machine learning, causal inference (Double ML), clustering, classification, anomaly detection, Bayesian statistics, regression analysis, and complex time-series modeling. Proven ability to translate advanced analytics into actionable insights for policy and decision-making.

Technical Skills

Programming: Python | R | SQL | SAS | Stata

ML/Stats: GLMs | Deep Learning (CNN | RNN | ANN) | Time Series | Causal Inference | Bayesian Methods

Tools: TensorFlow | PyTorch | Keras | Tableau | Power BI | LaTeX

Cloud: AWS | GCP | Snowflake

Education

PhD in Statistics University of Regina, Canada	2024
Certified Senior Data Scientist DASCA, USA	2019
MSc in Mathematics University of Mannheim, Germany	2016
BSc in Mathematics University of Mannheim, Germany	2013

Awards

German Government Program on Education of Youth | 2009—2016
Faculty of Graduate Studies and Research of U of Regina: Women in STEM Scholarship | 2019
MITACS Research Award | 2020 — 2021
University of Regina Research Award | 2021
Department of Mathematics and Statistics Research Assistantship | 2022
University of Regina Graduate Scholarship | 2022
Saskatchewan Innovation and Excellence Scholarship | 2020 — 2023
CRDCN Emerging Scholars Grant | 2024
SSHRC IG Grant | 2024—2026

Experience

Data Scientist

Canadian Centre of
Recovery Excellence
(CoRE)
Calgary, AB
2025 – Present

- Conduct advanced statistical modeling and inference on large-scale mental health and addictions datasets, including propensity score matching, logistic regression, multiple imputation, difference-in-differences, and multi-group interrupted time series analysis using Python and R.
- Design and execute A/B testing frameworks and statistical power analyses to quantify intervention effects and optimize study design.
- Perform inter-rater reliability analysis using intraclass correlation coefficients (ICC) to ensure consistency and validity of multi-observer data.
- Develop predictive and causal inference models integrating machine learning and econometric methods to identify determinants of treatment outcomes.
- Build automated data pipelines via API integration and web scraping to support real-time data ingestion and reproducible analytics.

SSHRC Postdoctoral Fellow

University of Regina
2024 – 2026

- Conducting SSHRC-supported research on AI-driven agricultural resilience, focusing on financial inclusion, climate and price risk forecasting, and hedging strategies for Canadian farmers.
- Leading a study on weather risk management, developing and evaluating weather derivatives as financial instruments for agricultural producers.
- Designing and implementing predictive and econometric models integrating financial, climatic, and behavioral data to assess risk exposure and resilience.
- Collaborating with faculty experts to prepare a SSHRC Insight Grant application and co-author academic publications in agricultural finance and risk management.

Mathematical Modeling Consultant

Farm Credit Canada
FCC
2022 – 2025

- Developed machine learning models using Python and R, leveraging libraries like pandas, scikit-learn, TensorFlow, LangChain, and PyTorch for regression, propensity propensity scoring, segmentation retention and time series forecasting.
- Provided strategic recommendations to senior management, ensuring evidence-based policy decisions through advanced analytics and data-driven insights.
- Effectively communicated complex technical concepts to technical and non-technical audiences, presenting insights and recommendations to senior leadership.
- Conducted Monte Carlo simulations and back-testing to validate models that informed strategic service planning.
- Delivered insights through customized dashboards and visual storytelling, helping leadership teams translate technical findings into actionable decisions.
- Mentored junior analysts and facilitated workshops to share analytical best practices and promote data literacy across departments.

Statistician (MITACS)

University of Regina
2021 – 2022

- Led national research on Indigenous agricultural business systems.
- Designed surveys, focus groups, and stakeholder engagement strategies.
- Delivered policy recommendations and co-authored a national report.